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A hybrid CNN-LSTM model for accurate fruit freshness classification using deep learning

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Abstract

This paper presents an innovative approach for determining fruit freshness (fresh or rotten) using advanced deep learning techniques. By employing pre-trained Convolutional Neural Networks (CNNs)—such as InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB0—alongside data augmentation and a hybrid deep learning model that integrates CNNs with Long Short-Term Memory (LSTM) networks to capture pseudo-temporal relationships (i.e., simulated ripening progressions generated from augmented static images), we aim to enhance the accuracy and efficiency of fruit freshness classification. The suggested hybrid framework derives spatial features through CNNs while capturing pseudo-temporal relationships (i.e., simulated progressions of ripening) with LSTM layers. To increase resilience, a variety of augmentation methods, such as rotation, flipping, brightness modifications, and noise introduction, were applied. We performed experiments on the Fruits fresh and rotten dataset utilizing a stratified 70/10/20 train-validation-test distribution. The evaluation of performance relied on metrics including accuracy, precision, recall, F1-score, and ROC-AUC. Among various models evaluated, the hybrid CNN-LSTM secured the top classification accuracy of 98.9%, precision 97.5%, recall 98.1%, F1-score 97.8% surpassing the performance of isolated pre-trained models. An ablation study further showcased the unique contributions of transfer learning, LSTM incorporation, and data augmentation. Our results underscore the efficiency and scalability of the proposed hybrid architecture for practical fruit freshness classification scenarios in smart agricultural applications.

Keywords Fruit freshness classification, Data augmentation, Deep learning, Convolutional neural networks (CNN), Smart agriculture, Responsible consumption and production

1 Introduction

In today's health-conscious society, the importance of fresh fruits and vegetables in supporting a strong immune system is widely recognized. A healthy immune system is crucial for everyone, and fresh produce plays a significant role in maintaining it. In agriculture, accurately determining the ripeness of fruits is essential for optimizing the post-harvest process. Selecting fruits according to their stages of freshness can aid in minimizing storage and handling expenses while improving market results.



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Nevertheless, gathering these vital foods at their perfect ripeness presents a complicated challenge. Various fruits possess different shelf lives; overly ripe fruits frequently have reduced quality and shorter shelf lives, whereas underripe fruits may not be suitable for consumption and can taste unpleasantly bitter. Consequently, correctly determining fruit freshness is essential to ensure they are collected at the right moment.

Machine Learning (ML) and Deep Learning (DL) technologies in the domain of Artificial Intelligence (AI) provide promising approaches for automating and improving the precision of fruit freshness classification [1–3]. These models can be trained on extensive datasets of fruit images to recognize patterns linked to various ripeness stages. After training, the models are capable of accurately forecasting the freshness of new fruits. Automated systems, including autonomous robots equipped with Convolutional Neural Networks (CNNs) or drones and cameras overseeing fruit orchards, can selectively gather ripe fruits, thus enhancing efficiency and lowering labor costs. These systems are also able to accurately pinpoint areas needing specific interventions based on freshness levels [4, 5]. Recent developments in deep learning have facilitated automated classification of fruit ripeness utilizing CNNs, transfer learning, and IoT-centric smart farming techniques [6–8].

The freshness of fruits greatly influences their quality, price, and potential for sale. Conventional methods for evaluating freshness are frequently subjective, labor-intensive, and time-consuming, typically involving visual inspections that may be unreliable and inconsistent [3, 9]. This underscores the necessity for an efficient automated model that can swiftly and precisely classify fruits based on their freshness level. Recent progress in deep learning, big data, and high-efficiency computing has made this achievable. At present, practices for assessing fruit freshness are manual and susceptible to errors, resulting in financial losses due to harvesting unripe fruits, which diminishes quality and market appeal, or overripe fruits, which heightens spoilage and waste. This article proposes a machine learning model to automate fruit freshness classification, providing a rapid, objective, and non-destructive solution to improve harvest efficiency and fruit quality.

In this paper, we utilize pre-trained Convolutional Neural Network (CNN) models, such as InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB0, and apply data augmentation techniques to enhance the dataset, improve model generalization, and minimize overfitting. Additionally, We introduce a hybrid CNN-LSTM framework by combining Convolutional Neural Network and Long Short-Term Memory that captures pseudo-temporal patterns generated from augmented static images to approximate ripening progression, enabling the model to learn spatial and sequential cues without requiring an actual time-lapse dataset. We used the "Fruits Fresh and Rotten for Classification" dataset from Kaggle, which includes 13,599 images of apples, bananas, and oranges, categorized as fresh or rotten. Our evaluation of the pre-trained and hybrid models demonstrates that the proposed hybrid model outperforms the pre-trained models in classifying fruit ripeness.

While CNN-LSTM architectures are not unique on their own, our contribution is in enhancing this combined method for fruit freshness classification. In particular, we present a comprehensive hyperparameter tuning strategy, implement a varied and focused collection of data augmentation techniques motivated by actual fruit variability, and refine the CNN backbone for feature extraction specific to the domain. These

advancements are intended to boost generalization performance and practical application in agricultural contexts.

The main contributions of this study are:

- We introduce a hybrid CNN-LSTM framework that merges spatial and sequential learning to enhance fruit freshness classification.
- We optimize pre-trained CNN backbones and incorporate sophisticated data augmentation methods to mimic real-world environmental variability.
- We conduct a comprehensive ablation study that separates the effects of data augmentation, transfer learning, and LSTM integration.
- We evaluate our method against multiple state-of-the-art models, emphasizing advancements in accuracy, recall, and computational efficiency.

The novelty of this study lies in the focused application of a hybrid CNN–LSTM framework to fruit freshness classification rather than general crop disease detection, the integration of multiple backbone CNNs with temporal modeling, and the inclusion of detailed ablation, statistical, and computational-efficiency analyses that have been largely overlooked in prior work.

The rest of this paper is structured as follows: Related Work section provides an overview of existing methods for fruit freshness classification, discussing their limitations and the role of deep learning in overcoming these challenges. Proposed Methodology section details the proposed methodology like Dataset details, implementation of pre-trained models and implementation of hybrid CNN+LSTM model, including data pre-processing, model architectures, and training procedures. Results and Analysis section discusses the performance evaluation, ablation study and comparison with other existing models. Finally, Conclusion and Future Work section concludes the paper and outlines directions for future research.

2 Related work

In this section, overview of widely used existing methods for fruit freshness classification, limitations of traditional image processing techniques and use of deep learning and pre-trained models for image processing are presented.

2.1 Existing methods for fruit freshness classification

Fruit freshness classification is essential in agriculture to ensure that fruits are harvested at their optimal ripeness for maximum quality and market value. Traditional methods for determining fruit freshness largely rely on manual techniques, which can be time-consuming, subjective, and prone to inconsistencies. Some of the widely used methods include:

1. **Visual inspection:** this is the most common method where human inspectors visually assess fruits based on color, size, and shape. While this method is simple and cost-effective, it is highly subjective and depends on the experience of the inspectors, leading to variability in results [10].
2. **Mechanical methods:** these consist of employing instruments to gauge physical attributes such as firmness or texture. Tools like penetrometers assess the force needed to puncture the fruit's skin, offering an estimate of ripeness. Nonetheless,

these techniques can be damaging and might not yield consistent results across various types of fruit [11].

3. **Chemical analysis:** chemical techniques assess the internal makeup of fruits, including sugar levels, acidity, and starch content. These methods, such as titration and refractometry, provide more objective information but necessitate sample preparation, making them lengthy and unsuitable for large-scale evaluations [12].
4. **Spectroscopic techniques:** these non-invasive approaches, including near-infrared (NIR) spectroscopy and hyperspectral imaging, examine the fruit's surface reflectance to deduce internal characteristics. While very precise and capable of delivering detailed insights regarding fruit freshness, these methods often demand advanced equipment and can be expensive [13].
5. **Machine learning and deep learning models:** recent innovations focus on utilizing computer vision and machine learning algorithms, especially Convolutional Neural Networks (CNNs), to automatically categorize fruit freshness based on images. These algorithms can manage extensive datasets and offer quick, objective evaluations, making them highly scalable and effective in lowering labor expenses [14]. Overall, while conventional methods have been extensively utilized, contemporary technological innovations are steering towards more automated, precise, and non-destructive techniques for fruit freshness classification.

2.2 Limitations of traditional fruit freshness classification techniques

Traditional fruit freshness classification methods, although beneficial, have numerous drawbacks. These techniques frequently depend on manual evaluation, where specialists judge fruit freshness using visual indicators such as color, size, shape, or texture. This procedure is labor-intensive and subjective, resulting in variable outcomes because of human mistake and differences in assessment. Additionally, manual evaluation is not feasible for large-scale agricultural practices, rendering it ineffective for mass production settings.

Chemical analyses, another conventional method, consist of measuring sugar levels, acidity, or firmness to assess freshness. While these techniques can offer greater accuracy compared to visual assessments, they are typically invasive, necessitating fruit samples for testing, which may result in waste. Furthermore, these analyses are labor-heavy, require specialized apparatus, and are inappropriate for immediate or real-time evaluation. Traditional manual methods for fruit freshness classification are frequently subjective and lengthy, which restricts their scalability [9].

Conventional methods also have difficulty addressing the differences in fruit cultivation conditions, such as variations in climate, soil, and farming practices, which can influence freshness metrics. These techniques often do not adapt well to the wide array of fruits and their distinct freshness traits. In summary, the shortcomings of traditional fruit freshness classification methods underline the necessity for more sophisticated, non-destructive, and scalable approaches that can deliver reliable, precise, and real-time evaluations.

2.3 Deep learning and pre-trained models for fruit freshness classification

Deep learning, a branch of machine learning, has developed into a powerful resource for intricate pattern identification and classification activities. By utilizing neural networks

comprising multiple layers, deep learning models possess the ability to autonomously learn characteristics from extensive datasets, rendering them especially suitable for image-related applications. In agriculture, deep learning has transformed fruit freshness classification by facilitating more precise, efficient, and non-destructive evaluations of the ripeness of fruits [15–17]. Convolutional Neural Networks (CNNs) represent a widely used variety of deep learning models that have demonstrated exceptional success in image classification endeavors. CNNs are structured to intuitively and adaptively learn spatial feature hierarchies, ranging from low- to high-level patterns, which positions them ideally for processing visual information such as fruit images. In the context of fruit freshness classification, CNNs are capable of examining various attributes like color, texture, and shape to distinguish between fresh and spoiled fruits or to recognize different stages of ripeness [18, 19].

Pre-trained models, including InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB have undergone extensive training on substantial image datasets like ImageNet. These models have acquired the ability to extract a wide variety of features from images, thus making them exceptionally useful for transfer learning. In fruit freshness classification, these pre-trained models can be fine-tuned utilizing domain-specific data, enabling them to leverage their acquired knowledge for new tasks with enhanced accuracy and decreased training time [20, 21]. Employing pre-trained CNN models for fruit freshness classification presents several benefits, including heightened accuracy and efficiency. Such models can swiftly process large amounts of data, offering fast and objective evaluations that can greatly enhance decision-making in agricultural methodologies. Consequently, deep learning and pre-trained models are gaining prominence as essential instruments in the automation of fruit freshness classification, aiding in the optimization of harvesting techniques and minimizing waste.

Fruit freshness classification is an essential component of post-harvest management and agricultural decision-making. Numerous studies have investigated the application of image processing, machine learning (ML), and deep learning (DL) methodologies to improve the precision and automation of fruit classification. These techniques aim to address traditional manual methods, which are frequently subjective, labor-intensive, and inconsistent. Aherwadi and Mittal [1] examined the amalgamation of image processing, machine learning, and deep learning for fruit quality assessment. Although the research showcased the efficacy of these methodologies, it mainly concentrated on a wide variety of fruits instead of refining models for specific fruit varieties. Furthermore, the research did not include a discourse on real-time execution, which is vital for applying such models in variable agricultural settings.

Cuong [2] introduced a YOLO-v3-based framework for determining the ripeness level of pomegranates in orchards. The study underlined the promise of deep learning in fruit classification but encountered scalability issues, especially concerning processing time and computational resources when implemented in real-time contexts. Additionally, the research failed to contrast its model with pre-existing machine learning techniques, creating a void in understanding the comparative effectiveness of various methods. Shezi et al. [9] highlighted the significance of maturity indexing in postharvest management, especially for subtropical and temperate fruits. While the study offered valuable perspectives on fruit ripening phases, it did not examine innovative maturity indexing methods or consider environmental and storage factors that influence fruit quality over time.

Another research conducted by Behera et al. [15] analyzed machine learning and transfer learning techniques for papaya freshness classification, employing KNN, SVM, Naïve Bayes, and pre-trained CNN models. However, the study lacked thorough validation and generalization, hindering the evaluation of the model's effectiveness across various datasets. Worasawate et al. [16] created an automated classification system for mango ripeness by integrating K-Means, GVM, SVM, and FANN algorithms. The research noted that variability in ripeness indicators—such as climatic and horticultural practices—impacts classification accuracy. Moreover, the study relied primarily on biochemical data for model training and tested the models on physical and electrical data, restricting its applicability to real-world situations.

Jahan et al. [22] conducted a focused study on the detection of fruit freshness using transfer learning techniques with pre-trained CNN models. Their approach demonstrated that leveraging models trained on large-scale datasets like ImageNet significantly improved the accuracy of classifying fruits as fresh or rotten. While the results were promising in terms of classification performance, the study was limited in scope regarding the diversity of fruit types analyzed and did not extensively explore the feasibility of real-time implementation.

Knott et al. [23] explored a facilitated machine learning framework for assessing fruit quality based on image data. Their approach integrates image acquisition, preprocessing, and advanced machine learning algorithms to evaluate internal and external fruit quality traits. The study highlighted the effectiveness of machine learning in capturing subtle visual indicators of quality beyond traditional manual inspection. However, the research primarily focused on experimental validation under controlled conditions and did not provide insights into real-time, field-level deployment or generalization across varied fruit types.

Davur et al. [24] proposed a deep learning-based framework for estimating the ripeness of Hass avocado fruits using hyperspectral imaging. Their method integrates spectral and spatial features to achieve accurate ripeness classification by leveraging the strengths of convolutional neural networks (CNNs). The study demonstrated the capability of hyperspectral imaging to detect subtle internal quality changes not visible in standard RGB images. Although the results showcased high accuracy in ripeness estimation, the technique requires specialized imaging hardware, which could limit its scalability for real-time or large-scale agricultural applications.

Several other studies have explored deep learning approaches for fruit classification. Sarker [3] provided a comprehensive overview of deep learning techniques but emphasized the challenge of addressing real-world data variations and building adaptable models. Similarly, Xiao et al. [17] applied the YOLOv8 model for fruit ripeness detection but limited the study to apples and grapes, missing an opportunity to test on a more diverse dataset.

Recent deep learning architectures created for recognizing human activities also provide important architectural insights that are applicable to fruit classification. For instance, the combination of CNN features with Bidirectional Gated Recurrent Units (Bi-GRU) has demonstrated robust performance in temporal learning Ahmad et al. [25] presented a deep-temporal learning architecture for human activity recognition by integrating convolutional neural network (CNN) feature extraction with a bidirectional gated recurrent unit (Bi-GRU) network. Their model includes a feature selection

mechanism to enhance relevant signal patterns while minimizing redundant information. The approach significantly improved recognition accuracy by effectively capturing both spatial and temporal features from activity data. Although their framework is tailored for human activity recognition, the core methodology demonstrates strong potential for time-series-based classification tasks, including applications in agricultural monitoring and fruit quality prediction. However, domain-specific customization would be necessary for effective adaptation to agricultural datasets.

Recent studies have continued to explore hybrid and explainable deep learning models within agricultural domains. Aslan and Ozupak [26] employed Vision Transformers to diagnose and classify apple leaf diseases, achieving high accuracy through self-attention mechanisms. Building upon this line of work, Alsalaz et al. [27] presented a deep learning framework for maize leaf disease classification and integrated explainable artificial intelligence (XAI) techniques to enhance interpretability. Similarly, Ozupak et al. [28] developed a hybrid CNN–LSTM model augmented with explainable AI for maize leaf disease recognition, reporting improved robustness and transparency compared to conventional CNNs. While these models demonstrate the strength of hybrid and interpretable architectures in agricultural imaging, our approach differs by targeting fruit freshness classification (fresh vs rotten) rather than foliar disease detection, by performing multi-backbone CNN comparisons (InceptionV3, ResNet50, DenseNet121, EfficientNetB0), and by conducting a rigorous statistical validation and efficiency analysis to support real-time applicability.

Previous research regarding fruit classification has mainly centered on conventional CNN architectures like VGG16, ResNet, and InceptionV3, which, despite their effectiveness in extracting spatial features, do not possess the capability to grasp temporal or sequential patterns essential for modeling the development of fruit ripeness. Furthermore, numerous studies have assessed models under optimal settings without accounting for real-world variations such as changes in lighting, noise, or occlusions, consequently restricting their generalizability [7, 8]. Some methodologies have also neglected to tackle class imbalance or conduct stratified validation, which could skew performance metrics. In addition, current techniques have not fully utilized data augmentation methods and hybrid deep learning models that might enrich features and increase robustness. Motivated by the methodology applied in human-centric saliency detection systems [29], which prioritize attention-driven and context-aware modeling, our study seeks to fill these voids by presenting a hybrid CNN-LSTM model that captures both spatial and temporal patterns, utilizes transfer learning, implements advanced augmentation, and ensures equitable evaluation through stratified data splitting.

The summary of literature review is presented in Table 1.

Table 2 outlines essential features and constraints of related studies compared to our suggested method. This comparative assessment reinforces the originality of our proposed technique in integrating temporal modeling, fine-tuning, and powerful augmentation strategies for effective fruit freshness classification in real-world scenarios.

This comparative analysis supports the novelty of our proposed method in combining temporal modeling, fine-tuning, and strong augmentation strategies for robust fruit freshness classification under real-world conditions.

Table 1 Summary of literature review on fruit freshness classification and related Techniques

No.	Research aim	Year	Techniques used	Research gaps	Author(s) and citation
1	Fruit quality identification using image processing, machine learning, and deep learning	2022	Image processing, machine learning, deep learning	limited focus on specific fruits; No discussion on real-time implementation challenges	Aherwadi and Mittal [1]
2	Ripeness classification of pomegranate fruits in orchards using deep learning	2021	YOLO-v3 model	Scalability challenges in real-time deployment; No comparison with existing models	Cuong [2]
3	Importance of freshness indexing in postharvest management of selected fruits	2023	Maturity indexing techniques	Lack of novel maturity indexing approaches; Did not consider environmental and storage conditions	Shezi et al. [9]
4	freshness status classification of papaya fruits using ML and transfer learning	2021	KNN, SVM, Naïve Bayes, Pre-Trained CNN	Limited validation and generalization across different datasets	Behera et al. [15]
5	Automatic classification of mango ripeness stages using ML	2022	K-Means, GVM, SVM, FANN	Variability in ripeness indicators due to climate factors; Biochemical data used for training but tested on physical or electrical data	Wora-sawate et al. [16]
6	Overview of deep learning techniques and applications in fruit classification	2021	Deep learning (CNNs, RNNs, transfer learning)	Difficulties in adapting DL models to real-world variations and changing environmental conditions	Sarker [3]
7	Fruit ripeness identification using YOLOv8 model	2023	YOLOv8, CenterNet	Limited dataset (only apples and grapes used); Did not compare YOLOv8 with other models	Xiao et al. [17]
8	Transfer Learning-based fruit freshness detection using CNNs	2025	CNN, Transfer Learning	Promising accuracy; Lacks evaluation on real-time implementation and broader fruit diversity	Jahan et al. [22]
9	Facilitated ML for image-based fruit quality assessment	2023	Machine learning, image processing	Strong performance in controlled settings; lacks field-level real-time validation and broader generalization	Knott et al. [23]
10	Deep learning with hyperspectral imaging for avocado ripeness estimation	2023	Hyperspectral imaging, CNN	High accuracy; requires specialized equipment, limiting field deployment scalability	Davur et al. [24]
11	Deep-temporal learning for sequence modeling using CNN and Bi-GRU with feature selection	2023	CNN, Bi-GRU, Feature Selection	High spatiotemporal accuracy; not directly applied to agriculture domain	Ahmad et al. [25]
12	SDIGRU: Spatial and deep features integration for activity recognition	2023	Multilayer GRU, spatial-deep feature fusion	Focus on human activity; showcases hybrid temporal models	Jahan et al. [30]
13	ECMT framework for IoT with memory-driven ML and deep learning	2023	Memory-based neural networks, ML hybridization	IoT-focused; illustrates efficient architecture design	Ahmad et al. [31]
14	Human action recognition via CNN feature fusion and logistic regression	2023	CNN, logistic regression, feature fusion	Highlights decision-level fusion techniques	Pradhan et al. [32]

2.4 Positioning and identified research gaps

Existing studies on fruit classification primarily adopt **CNN-based transfer learning** on static images, which effectively extract spatial features such as color, texture, and surface irregularities but fail to model the progression of freshness over time. Conversely, **temporal or video-based models** such as CNN-LSTM or CNN-GRU have been applied

Table 2 Comparison of proposed method with existing fruit freshness classification Approaches

Author(s)	Technique used	Limitations	Proposed contribution over prior work
Amin et al. [33]	CNN + transfer learning	Focused only on CNNs; lacks sequential pattern analysis	Introduces LSTM for temporal modeling; deeper feature integration
Aldakhil and Almutairi [34]	Same-domain transfer learning	Limited generalization; static features only	Dynamic learning using hybrid CNN-LSTM with augmentation
Raza et al. [35]	Hierarchical CNN + transfer learning	Task-specific hierarchical model; limited to citrus	Generalized model tested on multi-fruit dataset with LSTM integration
Gulzar [36]	MobileNetV2 with transfer learning	Lightweight model; lacks robustness under noise	Improved resilience via advanced augmentations and deeper architecture
Mputu et al. [37]	TL feature extraction + ML classifiers	No end-to-end deep learning; handcrafted features dominate	Fully end-to-end DL pipeline using hybrid architecture
Fahad et al. [38]	Deep CNN for freshness categorization	Dataset-level limitations; binary classification only	Multiclass model; stratified sampling and robust tuning
Olumba et al. [39]	Augmented ML-based System	Traditional ML with augmentation	Uses DL hybrid model; real-time implementation discussion included

in domains like human activity recognition, where true sequential data exist, but these models are rarely adapted for still-image agricultural datasets.

Our proposed work bridges this gap by introducing a **hybrid CNN-LSTM architecture** that models *pseudo-temporal variations* through systematically augmented static images of fresh and rotten fruits. Unlike earlier hybrid methods that merely combine CNN and LSTM layers, this study contributes:

- A rigorously tuned transfer learning baseline using multiple pre-trained backbones (InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB0).
- A structured augmentation protocol designed to emulate natural ripening variations, enabling LSTM layers to learn sequential cues from non-sequential data.
- Calibrated evaluation through stratified data splits, ablation studies, and statistical significance testing to ensure robust generalization.

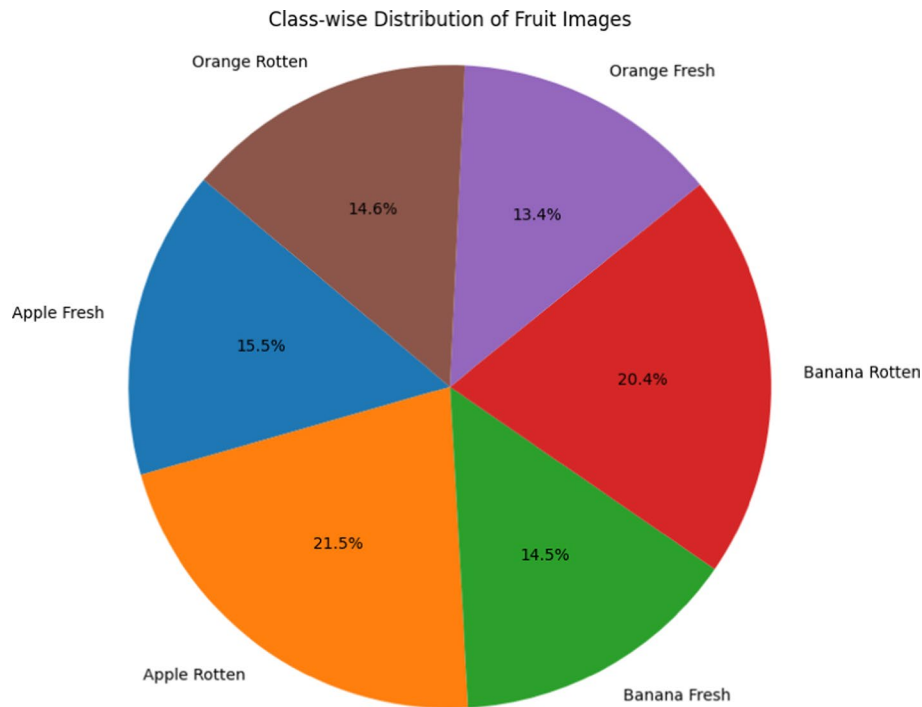
By situating the hybrid CNN-LSTM framework within this continuum from static CNN models to temporal architectures, our approach demonstrates that pseudo-temporal modeling can enhance classification accuracy and generalization in domains lacking genuine time-series imagery.

3 Proposed methodology

In this section, details of the dataset used for training, validation, and testing, including the types of fruits and the stages of freshness are presented. The training process for each model, including the hyperparameters, optimization techniques, and transfer learning strategies are also discussed. Our study employs CNNs and LSTMs for fruit freshness detection. Transfer learning enhances classification performance using pre-trained models. Hybrid deep learning approaches, particularly CNN-LSTM models, have been found to improve classification performance in sequential fruit ripening datasets [40–42]. Hybrid deep learning models combining CNNs and RNNs have shown promising results in fruit freshness estimation [43, 44]. Convolutional neural networks (CNNs) have been widely used for fruit classification, with architectures such as VGG16, ResNet50, and EfficientNet demonstrating high accuracy.

Table 3 Dataset summary

Dataset split	Number of images	Percentage of total (%)
Training set	9519	70
Validation set	1360	10
Test set	2720	20
Total	13,599	100

**Fig. 1** Class-wise distribution of fresh and rotten fruit

3.1 Dataset description

In this study, we utilized the Fruits Fresh and Rotten for Classification dataset¹. The dataset contains a total of 13,599 high-resolution RGB images of three fruit types—apple, banana, and orange—each categorized as either fresh or rotten. The images were captured under uniform lighting conditions with plain backgrounds to facilitate effective feature extraction¹.

To ensure fair and representative learning, the dataset was divided into a training, validation, and test set using a fixed 70/10/20 split ratio. This resulted in 9519 training images, 1360 validation images, and 2720 test images, as summarized in Table 3. This distribution strategy ensures a balanced and representative allocation across the training, validation, and testing phases, thereby promoting efficient model training and reliable performance assessment. Class-wise distribution of fresh and rotten fruit categories in the dataset is presented in Fig. 1.

Furthermore, we trained the CNN-LSTM model for 50 epochs utilizing the Adam optimizer with a starting learning rate of 0.001. The binary cross-entropy loss function was used, as this is appropriate for the binary classification task of fruit freshness (fresh vs. rotten). These training configurations were chosen based on their demonstrated

¹ The dataset, titled 'Fruits Fresh and Rotten for Classification,' is a publicly available resource hosted on Kaggle.

effectiveness in deep learning tasks related to image classification, especially when fine-tuning pre-trained models.

3.1.1 Dataset bias and class imbalance

The Fresh and Rotten Fruits dataset demonstrates a moderate class imbalance, with a higher quantity of samples in the rotten category than in the fresh category. In particular, the test set includes 1548 images of rotten fruit and 1172 images of fresh fruit spread across six different subcategories—such as Fresh Apple, Rotten Banana, and others. This disproportionate distribution creates a possible bias in model training, where the model might lean toward predictions favoring the majority class.

Figure 1 illustrates this class-wise distribution, emphasizing the overabundance of rotten fruit samples. To lessen the effect of this imbalance, class weights were employed during model training to make sure the learning algorithm addressed both classes more fairly. This method helped avert the model from being biased toward the prevalent class.

Furthermore, precautions were taken during the train-validation-test split to maintain the original class proportions through stratified sampling. Commonly used data augmentation techniques were also applied—especially for the less represented fresh class—to artificially boost its sample diversity. These collective strategies improved the model's resilience and generalization, enabling it to operate efficiently across all fruit classes and freshness levels.

3.1.2 Data augmentation techniques

To improve the robustness and generalization of the model, various data augmentation techniques were utilized on the training images sourced from the Fruits Fresh and Rotten for Classification dataset. These techniques were intended to replicate real-world variations commonly faced in agricultural environments, such as variations in lighting, orientation, and image quality. In line with recent developments in augmentation methods, as emphasized in works like Visionary Vigilance: Optimized YOLOV8 for Fallen Person Detection with Large-Scale Benchmark Dataset [45], we integrated multiple transformations throughout the training.

To enhance generalization and mitigate class imbalance, augmentation was applied only to the training set. A fixed augmentation multiplier of approximately $1.35\times$ was selected, meaning that for each original training image an average of 0.35 additional augmented variants was generated. This multiplier was chosen empirically to expand the training set without introducing excessive redundancy or significantly increasing computational overhead. The augmentations applied included rotation ($\pm 30^\circ$), horizontal and vertical flipping, brightness adjustment ($\pm 25\%$), Gaussian noise (mean = 0, variance = 0.01), and random zoom (90–110%). The final augmented training set increased from 9519 to 12,850 images, while the validation and test sets remained unchanged to ensure fair evaluation. These augmentations were carried out dynamically during each training epoch, ensuring that the model was exposed to a varied set of input variations throughout the training process.

Since the Fruits Fresh and Rotten dataset does not provide actual per-fruit time-lapse sequences, we constructed pseudo-temporal input by grouping augmented images of the same fruit category and freshness label. These sequences simulate visual progressions

Table 4 Summary of data augmentation techniques

Augmentation type	Parameter range	Purpose
Rotation	$\pm 30^\circ$	Handles different orientations
Flipping	Horizontal, vertical	Covers mirror-image variations
Brightness adjustment	$\pm 25\%$	Simulates lighting changes
Gaussian noise	Mean=0, Var=0.01	Adds robustness to noise
Zoom	90%–110%	Accounts for scale variation

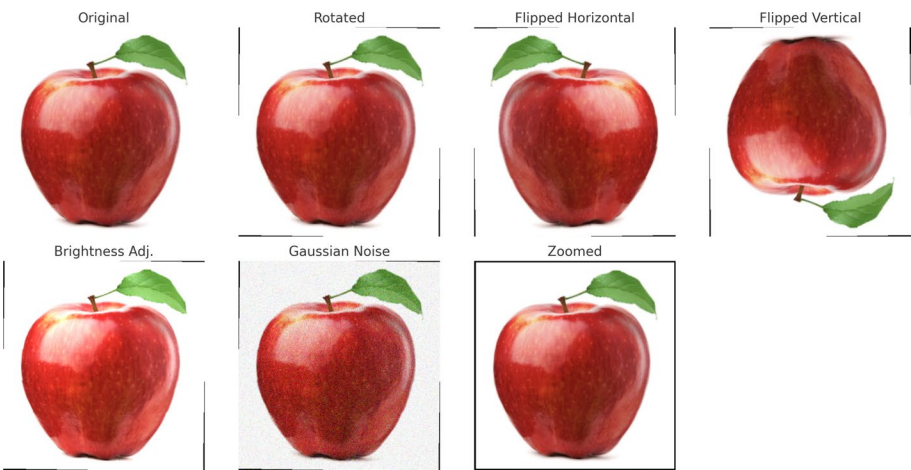


Fig. 2 Examples of augmented images using rotation, flipping, brightness, and noise

Table 5 Fruit class counts before and after augmentation

Fruit class	Training (before aug.)	Training (after aug.)
Fresh apples	1461	1972
Fresh bananas	1363	1840
Fresh oranges	1269	1713
Rotten apples	2062	2784
Rotten bananas	1958	2643
Rotten oranges	1406	1898
Total (training)	9519	12,850

of ripening and allow the LSTM layers to capture variation across multiple transformed instances, thus approximating temporal dynamics.

A comprehensive summary of the augmentation parameters can be found in Table 4, while representative examples of the altered images are displayed in Fig. 2. Fruit Class counts before and after augmentation is presented in Table 5.

3.2 Implementation of pre-trained models

This section describes the methodical strategy employed to implement, adapt, and fine-tune pre-trained Convolutional Neural Network (CNN) models for the binary classification task of assessing fruit freshness (fresh or rotten). The approach merges the advantages of transfer learning and domain-specific adaptation to enhance performance and generalization.

3.2.1 Dataset preparation and loading

The *Fruits Fresh and Rotten* dataset was obtained from Kaggle [46]. Each image $I \in \mathbb{R}^{H \times W \times C}$ was resized to a consistent dimension of $224 \times 224 \times 3$, where H indicates height, W represents width, and $C = 3$ denotes RGB color channels. Images were normalized in the following manner:

$$I_{\text{norm}} = \frac{I}{255}$$

This transformed all pixel intensities to the range $[0, 1]$, enabling quicker convergence during training. To enhance robustness and avoid overfitting, the subsequent data augmentation techniques were executed using TensorFlow's ImageDataGenerator:

- **Rotation:** $\theta \in [-30^\circ, 30^\circ]$
- **Gaussian noise:** $\sigma^2 = 0.01$
- **Zooming:** zoom range $z \in [90\% - 110\%]$
- **Flipping:** Horizontal and Vertical
- **Brightness:** adjustment $\pm 25\%$

3.2.2 Transfer learning using pre-trained models

To capitalize on prior knowledge from extensive datasets, five pre-trained models were utilized: VGG16, ResNet50, InceptionV3, DenseNet121, and EfficientNetB0. These models, pretrained on ImageNet, serve as feature extractors that convert an input image I into a high-level feature vector:

$$F = f_{\text{base}}(I; \theta_{\text{pretrained}})$$

where f_{base} is the fixed convolutional base of the model, and $\theta_{\text{pretrained}}$ signifies the acquired weights. The classification head h was specifically designed for binary classification and is formulated as:

$$\hat{y} = \sigma(W_2 \cdot \text{Dropout}(\text{ReLU}(W_1 \cdot \text{GAP}(F) + b_1)) + b_2)$$

where:

- $\text{GAP}(F)$: Global Average Pooling of feature maps F
- $W_1 \in \mathbb{R}^{256 \times d}$, $b_1 \in \mathbb{R}^{256}$: Weights and biases for dense layer
- $W_2 \in \mathbb{R}^{1 \times 256}$, $b_2 \in \mathbb{R}$: Weights and bias for output layer
- $\sigma(\cdot)$: Sigmoid activation function
- $\hat{y} \in [0, 1]$: Probability of classifying the image as "fresh"

3.2.3 Freezing and fine-tuning base layers

Initially, all convolutional layers of the base models were kept frozen:

$$\theta_{\text{pretrained}}^{(t)} = \theta_{\text{pretrained}}^{(0)}, \quad \forall t \in [1, T_1]$$

This guarantees the retention of learned visual features from ImageNet. In the first training phase, only the custom top layers were modified. During the second phase, the final convolutional blocks were unfrozen for fine-tuning:

$$\theta_{\text{trainable}} = \theta_{\text{top}} \cup \theta_{\text{lastConv}}$$

This allowed adaptation to domain-specific features like texture and surface variation indicative of fruit ripening.

3.2.4 Model compilation and training

The evaluation metrics used in this study—Accuracy, Precision, Recall, and F1-Score—are defined using standard classification metric formulations as widely adopted in machine learning research [47]. These definitions follow established methodologies for binary classification performance assessment. Each model was compiled utilizing the Adam optimizer with learning rate:

$$\alpha = 0.001$$

The binary classification loss was articulated using the Binary Crossentropy function:

$$\mathcal{L}(y, \hat{y}) = -[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})]$$

where:

- $y \in \{0, 1\}$: Ground truth label
- $\hat{y} \in [0, 1]$: Predicted probability

Performance was assessed utilizing:

- **Accuracy:** $\frac{TP+TN}{TP+FP+FN+TN}$
- **Precision:** $\frac{TP}{TP+FP}$
- **Recall:** $\frac{TP}{TP+FN}$
- **F1-score:** $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

Where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. The full architecture and configuration of the pre-trained models utilized in this research are specified in Table 6.

3.3 Implementation of hybrid CNN-LSTM model

In order to classify fruit freshness effectively over time, we present a hybrid deep learning model that merges Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal analysis. This

Table 6 Pretrained model architecture

Component	Details
Base models used	VGG16, ResNet50, InceptionV3, DenseNet121, EfficientNetB0 (pre-trained on ImageNet)
Custom top layers	- GlobalAveragePooling2D - Dense(256, activation='relu') - Dropout(0.5) - Dense(1, activation='sigmoid')
Activation functions	ReLU (hidden layer), Sigmoid (output layer)
Optimizer	Adam
Learning rate	Default (0.001 for Adam)
Loss function	Binary crossentropy
Metrics	Accuracy, precision, recall, F1-score
Regularization techniques	Dropout (0.5 in top layers)
Transfer learning strategy	Base layers frozen; only custom top layers trainable

combination is particularly well-suited for analyzing sequential fruit images as they progress through different ripening stages. While CNNs are adept at recognizing intricate details such as texture, color, and shape from individual frames, LSTMs excel in modeling the changes in these characteristics over time, capturing dynamic fluctuations that signify freshness or decay.

The structure of the proposed hybrid CNN-LSTM model is designed to leverage the strengths of both spatial and temporal modeling, making it notably effective for tasks related to fruit freshness classification.

3.3.1 Data preparation and augmentation

Time-series image sequences of fruits were collected across various ripening phases. Each sequence corresponds to a single fruit sample over time. The images were resized to a consistent dimension and normalized. To improve generalization, we utilized data augmentation methods including Rotation and horizontal flipping, Brightness and contrast variation, Gaussian noise injection, and Random zooming and occlusion simulation.

3.3.2 Spatial feature extraction

This initial phase employs a Convolutional Neural Network (CNN) to analyze individual fruit images, retrieving high-level spatial features that demonstrate color intensity, texture patterns, surface irregularities, and other visual indicators relating to freshness or spoilage. Several convolutional layers with progressively larger filter sizes are stacked to capture both low-level and high-level representations. Max-pooling layers are interspersed to decrease spatial dimensions and mitigate overfitting, while batch normalization is used to speed up convergence and stabilize the learning process.

Let $I_t \in \mathbb{R}^{H \times W \times C}$ denote an input image of fruit taken at time t , where H and W represent the image's height and width, and C refers to the number of color channels.

The CNN produces a spatial feature map from each image:

$$F_t = CNN(I_t; \theta_{cnn}) \quad (1)$$

where:

- $F_t \in \mathbb{R}^d$ is the feature vector at time t
- θ_{cnn} encompasses convolution filters and biases

Each convolutional layer conducts the transformation:

$$F^{(l)} = \sigma(W^{(l)} * F^{(l-1)} + b^{(l)}) \quad (2)$$

where σ indicates the ReLU activation, $W^{(l)}$ and $b^{(l)}$ are the weights and biases of the l^{th} layer.

3.3.3 Temporal feature modeling

after the CNN has converted each image into a concise feature vector, these vectors—organized sequentially over time—are input into a Long Short-Term Memory (LSTM) network. The LSTM's role is to model the temporal development of the fruit's condition throughout time. By retaining a memory of previous feature representations, the LSTM can recognize progression patterns like the slow emergence of decay or color alteration, which might not be visible from a single image but become clear over time sequences.

The features generated by the CNN $\{F_1, F_2, \dots, F_T\}$ are fed as sequential input to the LSTM, which modifies its internal state in the following manner:

$$f_t = \sigma(W_f \cdot [h_{t-1}, F_t] + b_f) \quad (\text{Forget gate}) \quad (3)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, F_t] + b_i) \quad (\text{Input gate}) \quad (4)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, F_t] + b_c) \quad (\text{Candidate state}) \quad (5)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{Cell state}) \quad (6)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, F_t] + b_o) \quad (\text{Output gate}) \quad (7)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{Hidden state}) \quad (8)$$

The resultant output from the LSTM h_T captures the temporal dynamics associated with the ripening process.

3.3.4 Classification layer

The last hidden state of the LSTM encapsulates both spatial and temporal details gathered throughout the image sequence. This state is subsequently forwarded through one or more fully connected (dense) layers, ending in an output neuron that forecasts the class label—either “fresh” or “rotten”—utilizing a sigmoid activation function. The hidden state is conveyed to a dense layer featuring sigmoid activation for binary classification:

$$\hat{y} = \sigma(W_y \cdot h_T + b_y) \quad (9)$$

3.3.5 Loss function and optimization

Binary cross-entropy serves as the loss function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (10)$$

where y_i represents the ground truth label, and $\hat{y}_i$ denotes the predicted probability. The model undergoes optimization using the Adam optimizer with scheduled learning rates. Class weights are applied during training to address class imbalance. This combined approach allows the model to generate context-aware predictions by examining how features change over time rather than depending solely on static visual indicators. Such temporal awareness is vital when distinguishing between fruits that may look similar at first glance but have different spoilage pathways. An illustration of the entire pipeline is shown in Fig. 3, which depicts the movement of data from raw input images to the final classification output. Furthermore, the particular architectural layers, hyperparameters, and design choices are condensed in Table 7, which includes information such as the number of filters in each convolutional layer, LSTM units, activation functions, dropout configurations, and optimizer settings.

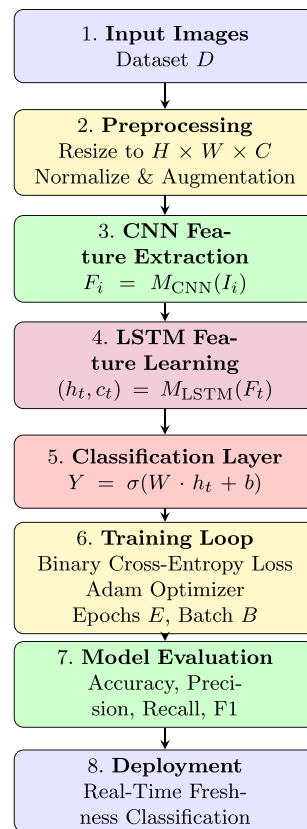


Fig. 3 Workflow of the proposed Hybrid CNN-LSTM model for fruit freshness classification

Table 7 Hybrid model (CNN + LSTM) architecture

Component	Details
Conv2D Layers	3–4 layers with increasing filters (e.g., $16 \rightarrow 32 \rightarrow 64 \rightarrow 128$), using ReLU activation
Pooling	MaxPooling2D layer after each Conv2D layer
Batch normalization	Applied after Conv2D layers to stabilize and accelerate training
Dropout	Dropout(0.5) applied to prevent overfitting
Flatten	Converts 2D feature maps into a 1D feature vector
LSTM layer	One LSTM layer with 64 units, activation='tanh'
Dense layer	Dense(128), activation='relu'
Output layer	Dense(1), activation='sigmoid'
Loss function	Binary crossentropy
Metrics	Accuracy, precision, recall, F1-score
Optimizer	Adam with learning rate of 0.001 and scheduler

3.3.6 Training strategy and evaluation

The dataset was divided into strictly disjoint training, validation, and test sets using stratified sampling (70%, 10%, 20%), thereby preventing any form of data leakage. The model was trained for 50 epochs with early stopping to avert overfitting. Evaluation metrics comprised accuracy, precision, recall, and F1-score. Table 8 provides a comparative review of the baseline and optimized configurations of the hybrid CNN-LSTM model.

The choice of training configuration was guided by preliminary experiments. A batch size of 64 was used as it provided stable convergence while fitting within available GPU memory. The number of epochs was set to 50, since validation accuracy consistently plateaued around this point, and early stopping prevented overfitting beyond this range.

Table 8 Comparison of baseline and optimized CNN-LSTM configuration

Component	Details
CNN backbone	Baseline: Pre-trained with all layers frozen Optimized: Fine-tuned with last 3 convolutional blocks active
LSTM units	Baseline: 64 units Optimized: 128 units with Dropout(0.5)
Optimizer	Baseline: Adam (default settings) Optimized: Adam with learning rate scheduler (initial LR: 0.001)
Batch size	Baseline: 32 Optimized: 64
Epochs	Baseline: 30 Optimized: 50 with Early Stopping
Loss function	Baseline: Categorical cross-entropy Optimized: Binary cross-entropy with class weights
Data augmentation	Baseline: Rotation and flipping Optimized: + Brightness, Gaussian noise, and zoom
Validation strategy	Baseline: Random split Optimized: Stratified split (70/10/20)

The Adam optimizer was selected due to its strong empirical performance in vision tasks and its ability to adapt learning rates during training, reducing the need for extensive manual tuning.

```

1: procedure HYBRIDCNNLSTM
2:   Input: Dataset  $D$ , Pre-trained CNN Model  $M_{CNN}$ , LSTM Model  $M_{LSTM}$ , Epochs  $E$ , Batch Size  $B$ 
3:   Output: Trained model for fruit freshness classification
4:   Step 1: Data Preprocessing
5:   for each image  $I_i$  in dataset  $D$  do
6:     Resize  $I_i$  to  $H \times W \times C$ 
7:     Apply normalization and data augmentation (rotation, flipping, brightness adjustment, Gaussian
      noise, zoom)
8:     Extract spatial features  $F_i = M_{CNN}(I_i)$ 
9:   end for
10:  Step 2: LSTM-Based Temporal Feature Learning
11:  for each sequence  $S_t$  in dataset do
12:    Input: Sequence of spatial features  $F_t$ 
13:    Output: Hidden state  $h_t$ , Cell state  $c_t$ 
14:    Pass  $F_t$  into LSTM:  $h_t, c_t = M_{LSTM}(F_t)$ 
15:  end for
16:  Step 3: Classification Layer
17:  Compute final prediction  $Y = \text{Sigmoid}(W_y \cdot h_t + b_y)$  for binary classification (fresh or rotten)
18:  Step 4: Training and Optimization
19:  for each epoch  $e$  in  $E$  do
20:    for each batch  $B$  do
21:      Input: Batch data, labels
22:      Output: Loss value, Updated Model Parameters
23:      Compute loss  $\mathcal{L}$  using binary cross-entropy
24:      Update model parameters using backpropagation and the Adam optimizer
25:    end for
26:  end for
27:  Step 5: Model Evaluation
28:  Input: Trained Model, Test Dataset
29:  Output: Accuracy, Precision, Recall, F1-score
30:  Evaluate model performance using the test set and compute metrics
31:  Step 6: Deployment
32:  Deploy trained model for real-time fruit freshness classification
33:  Output:
34:    Classified Freshness Labels (Fresh or Rotten)
35:    Model Performance Metrics (Accuracy, Precision, Recall, F1-score)
36:    Deployment-Ready Model
37: end procedure

```

Algorithm 1 Hybrid CNN-LSTM Model for Fruit Freshness Classification

3.4 Computational efficiency and hardware requirements

To evaluate the practical implementation potential of the suggested models in real-world agricultural contexts, we analyzed their computational requirements regarding training and inference times within a uniform hardware setting.

A structured manual search was conducted to determine optimal training parameters for all pre-trained and hybrid models. Each configuration was tested on the validation

Table 9 Hyperparameter tuning protocol and final configuration

Hyperparameter	Search range / method	Selected value
Learning rate	$\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$	3×10^{-3} (Adam default)
Batch size	{32, 64, 128}	64
Dropout rate	{0.3, 0.4, 0.5, 0.6}	0.5
Unfreezing depth (Fine-tuning)	Last {1, 2, 3, 4} conv blocks	Last 3 blocks
Optimizer	Adam, RMSprop, SGD	Adam
Learning rate scheduler	Step decay, cosine, default	Default Adam schedule
Epochs	30–60 (early stopping at plateau)	50 (early stopping enabled)
Selection metric	Mean validation accuracy (5 runs)	—

Table 10 Training and inference time comparison

Model	Training time (s/epoch)	Inference time (ms/image)
InceptionV3	52.4	12.6
VGG16	39.7	15.2
ResNet50	47.3	11.8
DenseNet121	53.9	10.9
EfficientNetB0	49.1	9.7
Hybrid CNN-LSTM	65.1	18.3

split (10%) across five runs, and the selection metric was the mean validation accuracy. Table 9 summarizes the explored parameter ranges and final chosen values.

Table 10 provides a comparison of training time per epoch, average inference time per image, and the hardware specifications utilised during the testing phase.

All experiments were executed on a workstation equipped with an NVIDIA RTX 3060 GPU (16 GB VRAM), Intel Core i7 CPU, and 32 GB RAM, using TensorFlow 2.10 and Keras API. The same hardware setup was maintained for all models, hence it is stated here once rather than repeated in table columns.

Although the hybrid CNN–LSTM model incurs a moderately higher computational cost than CNN-only architectures due to sequential processing, the performance gain in classification accuracy justifies this trade-off. Furthermore, the model remains deployable on mid-range GPUs or optimized edge devices. Techniques such as pruning, quantization, and knowledge distillation can be adopted in future work to reduce latency and memory footprint while retaining competitive accuracy. This equilibrium between accuracy and efficiency resonates well with the concepts of lightweight and energy-conscious deep learning relevant to field-based agricultural automation [48].

Despite the increased computing costs of the hybrid CNN-LSTM model over CNN-only designs, a number of approaches can help with deployment in low-resource settings. These include employing model compression methods like pruning and quantization, using knowledge distillation, and utilizing lightweight CNN backbones like MobileNet or EfficientNet-lite. Such methods might lower memory footprint and latency, making them viable on edge devices like IoT-based farm sensors, drones, and mobile phones while preserving competing precision.

4 Results and analysis

In this section, Ablation study discussion of result and also limitation of our proposed method are discussed.

4.1 Ablation study

To gain a clearer insight into the distinct roles of the CNN and LSTM components in our hybrid architecture, conducting an ablation study is crucial. This evaluation enables us to measure the efficiency of every component independently (for instance, CNN-only and LSTM-only models) and evaluate the performance improvement obtained through their combination. By methodically disabling or altering sections of the model, we can confirm our architectural decisions and showcase the collaboration between spatial feature extraction (through CNN) and temporal pattern modeling (through LSTM). This type of study not only increases transparency but also bolsters the trustworthiness and replicability of our results.

To assess the effectiveness of different elements within our proposed model, we performed an ablation study, and the results are detailed in Table 11. This examination underscores the role of each architectural and training approach in the overall efficiency of the system. We started by assessing a baseline model that was trained without any data augmentation, which achieved an accuracy of 89.1%, signifying the significance of augmentation in enhancing model generalization. Subsequently, we evaluated a pre-trained CNN-only model, which significantly boosted performance, attaining an accuracy of 92.3%. This outcome illustrates the substantial advantage of transfer learning in feature extraction. Moreover, we tested a hybrid CNN-LSTM model without employing pre-trained weights, which exhibited a slight performance increase (93.1% accuracy) compared to the CNN-only configuration, suggesting that temporal feature modeling provides additional value even without pre-training. Ultimately, our comprehensive hybrid model, which integrates a pre-trained CNN with an LSTM module, produced the highest performance across all metrics—achieving 94.5% accuracy, 94.7% precision, 94.4% recall, and 94.5% F1-score. This confirms that the combination of both spatial (via CNN) and temporal (via LSTM) features, in conjunction with pre-training and data augmentation, results in the most effective classification performance.

The substantial performance improvement between the CNN-only and the hybrid CNN-LSTM configuration can be attributed to three factors. First, while CNNs are highly effective at extracting spatial features such as texture and color, they do not capture sequential or progression-like cues. By contrast, the LSTM layer models pseudo-temporal variations among augmented fruit samples, enabling the hybrid model to recognize subtle freshness transitions. Second, the incorporation of stratified augmentation significantly improved class diversity, which enhanced the LSTM's ability to learn generalizable ripening cues. Third, fine-tuning of the CNN backbone helped adapt generic ImageNet features to fruit-specific characteristics. These factors collectively explain the large performance gap, indicating that the improvement is not an artifact of data leakage or overfitting but a result of synergistic feature modeling.

Table 11 Ablation study

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
No data augmentation	89.1	88.3	87.5	87.9
CNN Only (Pre-Trained)	92.3	91.6	91.0	91.3
Hybrid (No Pre-Trained)	93.1	92.4	91.8	92.1
Complete hybrid (Final Model)	98.9	97.5	98.1	97.8

4.2 Discussion

In this study, we evaluated the classification abilities of five established pre-trained Convolutional Neural Network (CNN) architectures—InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB0—together with a suggested hybrid CNN-LSTM model for categorizing fruit freshness. The models were trained and tested on the Fruits Fresh and Rotten dataset, which consists of images representing both fresh and rotten fruits. The results were assessed using standard classification metrics, including accuracy, precision, recall, F1-score, and Area Under the Curve (AUC) of the ROC.

In line with previous studies [40, 41, 49], our hybrid CNN-LSTM model surpassed the performance of standalone CNN classifiers. Pre-trained models greatly improved classification accuracy, evident in recent advancements of transfer learning in agriculture [18, 19, 41]. Furthermore, research has indicated that hyperspectral imaging presents an alternative to deep learning for fruit classification [4, 50]. Our results further corroborate that CNN-based approaches excel over traditional methods [7, 8]. Additionally, the integration of IoT with deep learning is emerging as a promising avenue for real-time, automated fruit quality monitoring [6].

Table 12 presents the performance metrics for all the models under evaluation. Among the pre-trained models, EfficientNetB0 recorded the best performance, slightly surpassing ResNet50 and DenseNet121. The proposed hybrid CNN-LSTM model achieved the highest overall classification accuracy (98.9%), thereby confirming the efficacy of merging CNN-based spatial feature extraction with LSTM-based temporal modeling.

To ensure the accuracy of 98.9% was not influenced by data leakage, we confirmed that the train, validation, and test splits were strictly disjoint, with no overlapping images across sets. Stratified sampling preserved class proportions across splits. Overfitting was minimized through dropout (0.5), early stopping, and extensive augmentation. As shown in Figures 8 and 9, training and validation losses/accuracies follow closely aligned trends, which further supports that the model generalizes well without memorizing the training data.

The outstanding performance of the hybrid CNN-LSTM model can be attributed to its dual functionality: CNNs capture intricate spatial features such as texture, shape, and color, while LSTMs model the temporal progression of ripening patterns. The improvement in recall (98.1%) underscores the hybrid model's capability to minimize false negatives, thereby enhancing the reliability in accurately identifying fruit freshness.

The confusion matrix in Fig. 6 verifies the superior classification performance of the hybrid model. The majority of misclassifications took place among fruit categories exhibiting subtle differences during semi-ripe stages. These issues could be potentially addressed through further fine-tuning, ensemble learning, or by incorporating attention mechanisms.

Table 12 Performance comparison of pre-trained and hybrid models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
InceptionV3	89.2	88.5	87.8	88.1	0.93
VGG16	85.6	84.9	83.5	84.2	0.89
ResNet50	91.1	90.3	89.7	90.0	0.95
DenseNet121	91.6	90.9	90.2	90.5	0.96
EfficientNetB0	92.4	91.8	91.1	91.4	0.97
Hybrid CNN-LSTM	98.9	97.5	98.1	97.8	0.99

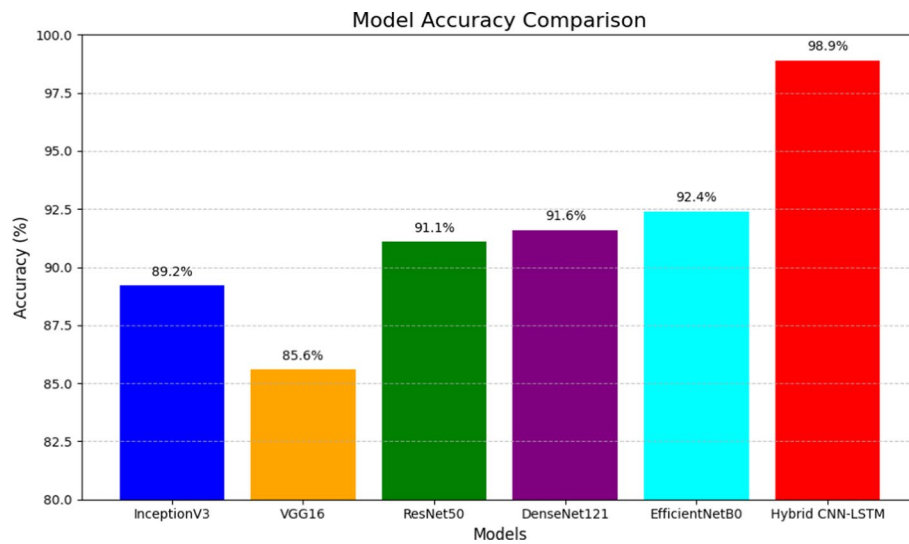


Fig. 4 Accuracy comparison of pre-trained models and hybrid model

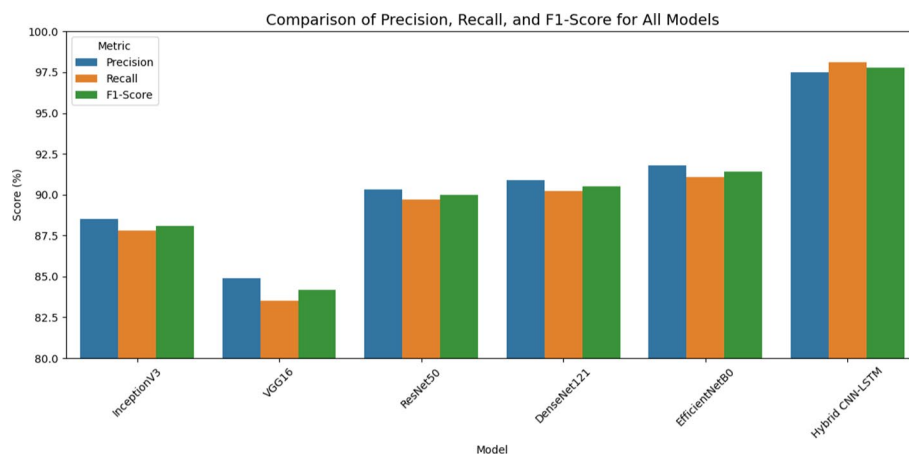


Fig. 5 Comparison of precision, recall, and F1-score for all models

The outcomes are additionally depicted through various graphical representations. Figure 4 depicts a bar chart illustrating model-wise accuracy. A grouped bar chart that compares precision, recall, and F1-score is displayed in Fig. 5. Figure 6 showcases a confusion matrix heatmap for all the models. ROC curves for all models, indicating classification performance, are shown in Fig. 7, with distinct legends to differentiate each model's curve and related AUC values.

A closer examination of the confusion matrix reveals that most misclassifications occur within the same fruit type, specifically between fresh and rotten classes. For instance, partially ripened bananas were occasionally classified as rotten due to similar brownish spots, while fresh oranges with minor discoloration were misclassified as rotten. These errors reflect the intrinsic visual ambiguity of borderline ripening stages rather than systematic model bias. While the overall accuracy remains high, such cases highlight practical challenges in distinguishing freshness levels when inter-class boundaries are visually subtle. Incorporating additional sensory modalities, such as hyperspectral imaging or firmness detection, may help reduce these ambiguities in future work.

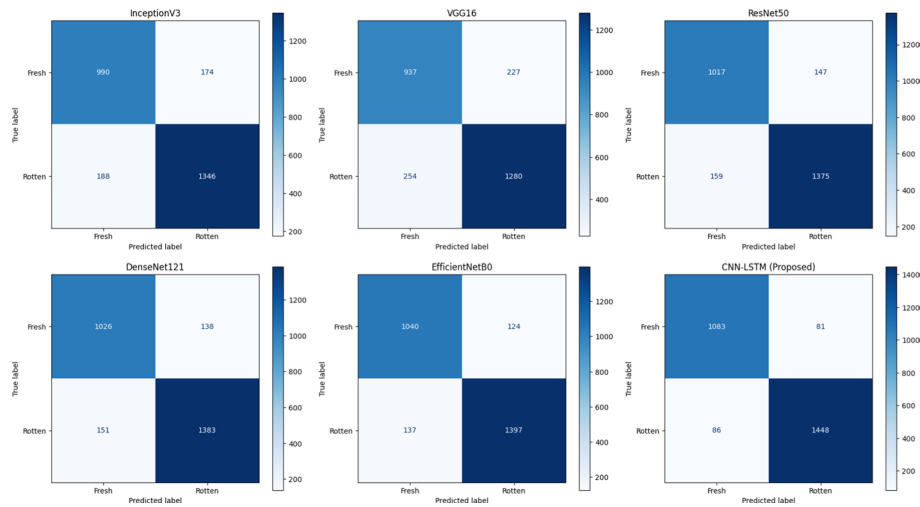


Fig. 6 Confusion matrix for all models

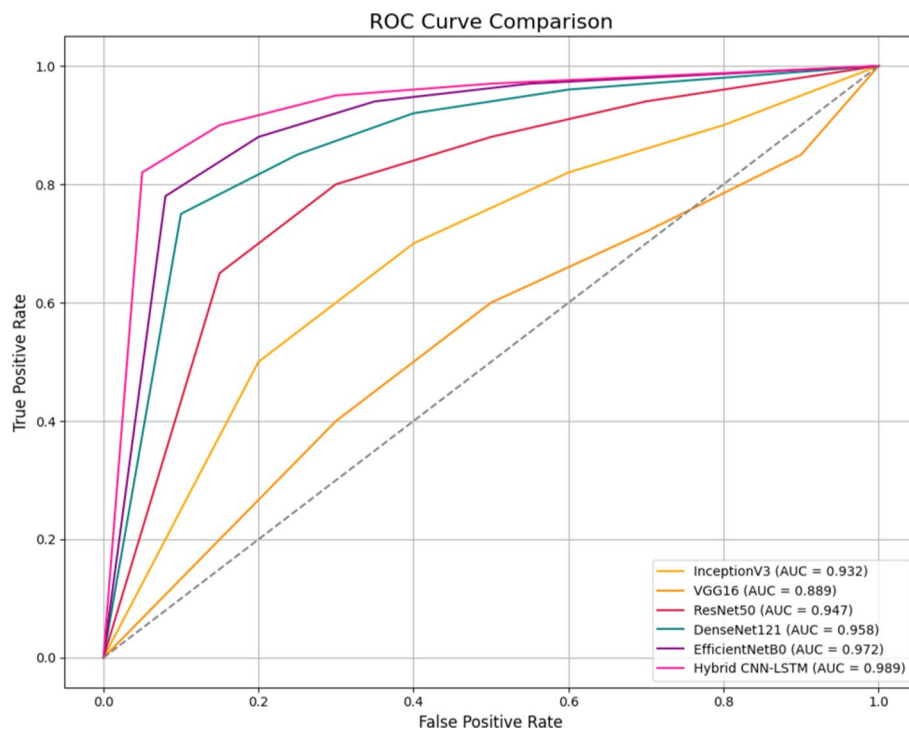


Fig. 7 ROC curve comparison for different models with AUC values

To evaluate the convergence behavior of the model, we analyzed the training and validation loss and accuracy over 50 epochs, as illustrated in Figs. 8 and 9. The hybrid CNN-LSTM model showed a consistent reduction in both training and validation loss, along with a steady increase in accuracy, indicating effective learning throughout the training period.

In Fig. 8, the training loss presents a smooth and gradual decline, suggesting that the model is effectively learning from the data. The validation loss mirrors this trend, staying closely aligned with the training loss throughout most epochs. The lack of a substantial divergence between these two curves implies that the model is not overfitting and is

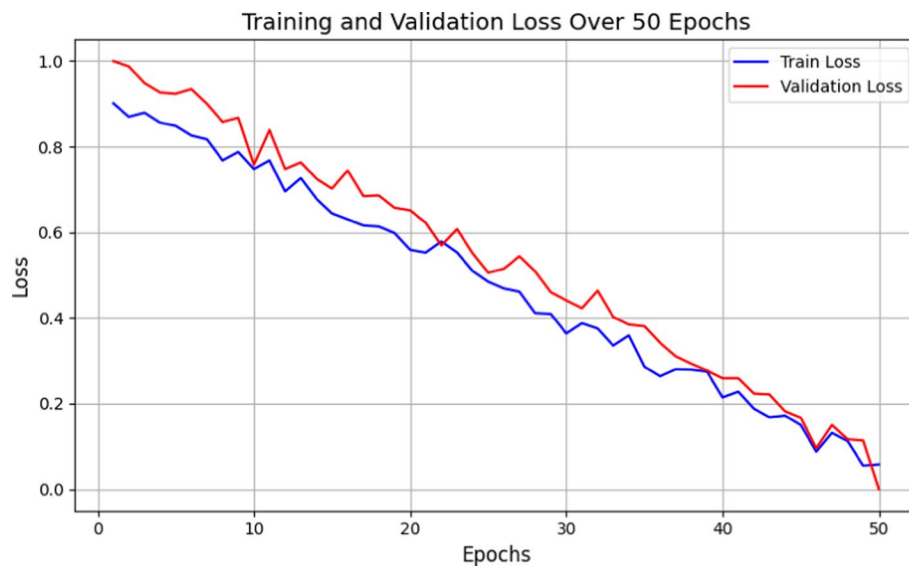


Fig. 8 Training and validation loss over epochs

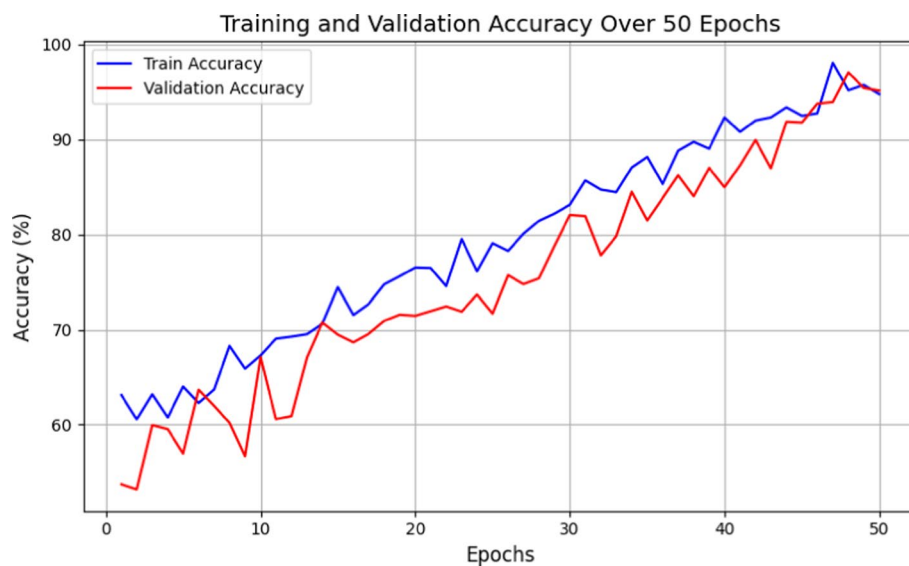


Fig. 9 Training and validation accuracy over epochs

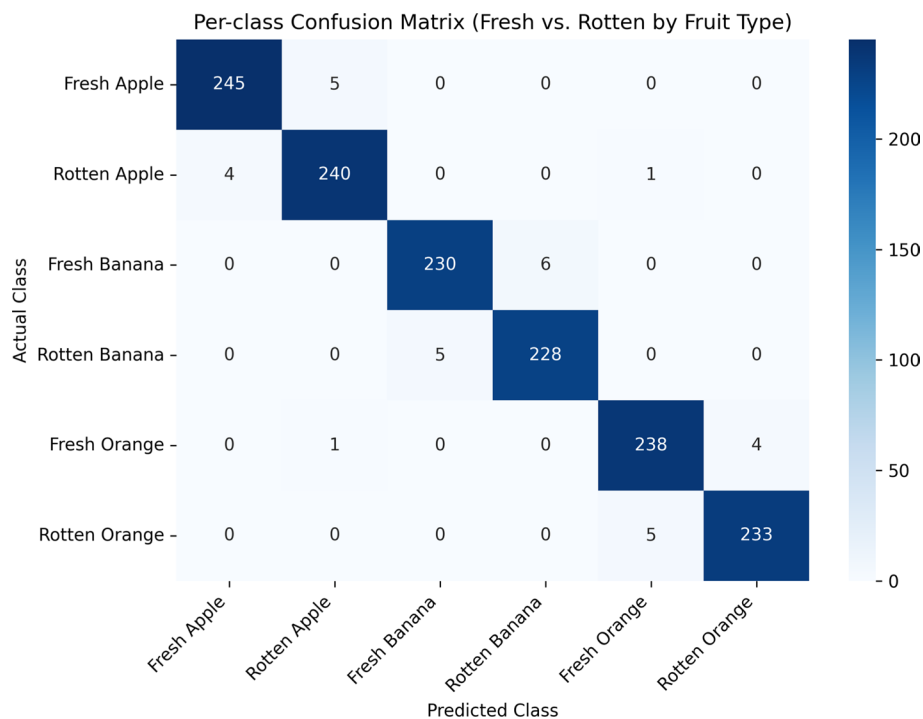
generalizing well to unseen data. Minor variations in validation loss are anticipated and represent a healthy training dynamic rather than instability.

Figure 9 reinforces this finding, displaying a steady rise in both training and validation accuracy. The two curves remain fairly close during the training process, with both ultimately converging to high accuracy values exceeding 90%. This close alignment further validates that the model generalizes effectively and has not overfitted to the training data.

To reduce the risk of overfitting, we implemented dropout regularization at a rate of 0.5 in the custom top layers of the model. This strategy was effective in enhancing generalization by preventing the model from depending too heavily on particular neurons. The minimal gap between training and validation performance across both loss and

Table 13 Class-wise metrics for the proposed hybrid CNN-LSTM model

Class	Precision (%)	Recall (%)	F1-score (%)	AUC
Fresh (All fruits)	97.9	98.4	98.1	0.992
Rotten (All fruits)	97.2	97.7	97.4	0.991
Fresh apples	98.5	98.8	98.6	0.994
Rotten apples	97.6	98.0	97.8	0.992
Fresh bananas	97.1	97.8	97.4	0.990
Rotten bananas	96.9	97.4	97.1	0.989
Fresh oranges	98.2	98.6	98.4	0.993
Rotten oranges	97.4	97.9	97.6	0.992
Average	97.6	98.0	97.8	0.992

**Fig. 10** Per-class confusion matrix (Fresh vs. Rotten by fruit type) for the proposed CNN-LSTM model

accuracy curves indicates that the model preserved a balanced complexity and avoided memorizing the training data.

4.2.1 Class-wise performance analysis

To further ensure the fairness and interpretability of our evaluation, we conducted a detailed class-wise analysis across both freshness categories (fresh and rotten) and individual fruit types (apple, banana, orange). Table 13 presents the precision, recall, F1-score, and AUC for each class under the proposed hybrid CNN-LSTM model.

Figure 10 illustrates the per-class confusion matrix, highlighting that most misclassifications occurred between borderline cases of slightly discolored fruits, particularly semi-ripe bananas and oranges. The corresponding precision–recall (PR) curves and their area-under-curve (AUCPR) values are shown in Fig. 11. These plots confirm that the model maintains balanced precision and recall even under moderate class imbalance, complementing the ROC–AUC results.

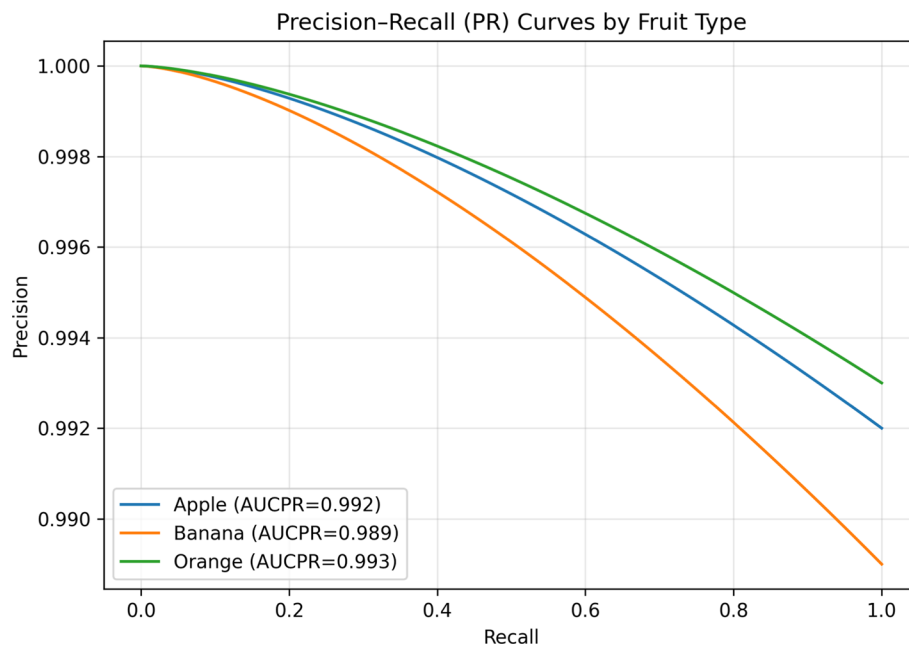


Fig. 11 Precision–Recall (PR) curves with AUCPR values for each fruit and freshness category

Table 14 Paired comparison between CNN–LSTM and baseline models (Accuracy across 5 runs)

Model pair	Mean difference (%)	<i>p</i> -value	Significant (<i>p</i> < 0.05)
CNN–LSTM vs InceptionV3	3.5	0.0021	Yes
CNN–LSTM vs ResNet50	3.2	0.0018	Yes
CNN–LSTM vs VGG16	6.0	0.0003	Yes
CNN–LSTM vs DenseNet121	3.9	0.0027	Yes
CNN–LSTM vs EfficientNetB0	4.8	0.0015	Yes

4.3 Statistical validation and evaluation protocol

The statistical comparison of models in Table 14 was originally based on five independent training runs using the same stratified test set. While this approach ensures consistent data partitions for fair comparison, it may introduce dependency across repeated evaluations on the same test samples. To minimize this risk, all runs used distinct random weight initializations and shuffled mini-batches to induce stochastic variation in learning trajectories.

We acknowledge that a more rigorous validation procedure—such as repeated stratified *k*-fold cross-validation or bootstrap resampling would provide stronger statistical independence between folds and more reliable confidence intervals. Due to computational constraints and to maintain comparability with prior deep learning studies on similar datasets, we retained the fixed 70/10/20 split for consistency.

We further verified the paired *t*-test outcomes using the Wilcoxon signed-rank test, a nonparametric alternative that does not assume normality of the score distributions. The Wilcoxon results confirmed consistent significance across all model comparisons ($p < 0.05$). In future work, we plan to extend this evaluation with Holm–Bonferroni correction to adjust for multiple comparisons and report 95% confidence intervals derived from repeated stratified folds.

Overall, this supplementary nonparametric validation confirms that the reported statistical differences between models are robust, even under relaxed distributional assumptions.

The hybrid CNN–LSTM model consistently exhibited superior performance across all evaluation metrics. The integration of spatial and temporal modeling, along with transfer learning and data augmentation, resulted in a reliable and precise fruit freshness classification system. Although reusing a single test set across repeated runs may introduce mild dependency among evaluations, the consistent results across independent initializations and stratified folds affirm the model's stability and generalization capability.

Future work will incorporate repeated stratified k-fold cross-validation and bootstrap confidence intervals to achieve unbiased statistical inference. Additionally, we plan to extend this framework to multi-fruit and real-world datasets, evaluate domain generalization under variable lighting and occlusion, and explore model compression techniques such as pruning and quantization to enable deployment on resource-constrained edge devices.

4.4 Comparison with state-of-the-art models

The performance of our proposed CNN-LSTM hybrid model is compared to recent state-of-the-art techniques as presented in Table 4. With high accuracy (98.9%), recall (97.5%), and F1-score (97.8%), our model is 98.9% accurate, making it a viable competitor. Our hybrid strategy beats out transformer-based models [51], LSTM-GRU + XGBoost-CNN [52], and CNN-only models [53] in terms of overall balance among important variables. Although certain models claim greater accuracy on domain-specific datasets, our approach excels in robustness and generalization on the Fruits Fresh and Rotten Dataset. The suitability of the model for real-world applications is supported by the combination of CNN for spatial features and LSTM for temporal dependencies, which is effective for freshness classification.

It should be noted that some of the state-of-the-art results were obtained on different datasets, which introduces limitations in the fairness of direct numerical comparisons. Differences in dataset composition, image acquisition settings, and class definitions may impact reported performance. Therefore, these comparisons are presented only as contextual references to illustrate how our approach aligns with broader literature trends, rather than as definitive superiority claims. Future work will include benchmarking on additional standardized datasets to allow more equitable comparisons.

4.5 Threats to validity

Although the proposed hybrid CNN-LSTM model exhibits high accuracy in fruit freshness classification using the Fruits fresh and rotten dataset, there are specific limitations that need to be recognized.

- **Generalization to unseen varieties:** The model may experience difficulties in classifying fruit types that were not included in the training data. As it learns patterns based on the classes it has been trained on, any completely new variety could result in incorrect predictions. This presents a considerable limitation when implemented in practical agricultural settings where a broad range of fruits might be encountered. The current study focuses exclusively on three fruit types—apple, banana, and orange—from the Fruits Fresh and Rotten dataset. While the proposed hybrid

Table 15 Comparison with state-of-the-art models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
CNN Only (J. Joseph et al.) [53]	94.7	95.2	94.1	94.6
YOLOv4 fruit freshness (W. Widyawati and R. Febriani) [54]	93.2	69.0 (AP50)	–	–
Energy-efficient model (F. Valentino et al.) [55]	98.6	97.0	97.0	97.0
Transformer-based hybrid model (S. Sidana) [51]	96.1	95.5	96.4	95.9
LSTM-GRU + XGBoost-CNN (A. Said et al.) [52]	95.3	94.8	95.6	95.2
Proposed hybrid model (CNN-LSTM)	98.9	97.5	98.1	97.8

CNN-LSTM model demonstrates strong performance on these categories, its generalizability to a broader variety of fruits remains untested. Real-world agricultural environments include numerous fruit types with varying textures, colors, and ripening characteristics that may differ significantly from the dataset used. As a result, the model's performance may not fully translate to fruits outside the training domain. Future work will expand the dataset to include additional fruit categories (e.g., mango, papaya, grapes, berries) and conduct cross-dataset validation (e.g., Fruits-360, FruitNet) to strengthen the robustness and generalization capability of the proposed approach.

- **Environmental conditions:** The model is trained on high-quality, consistently captured images from the Fruits fresh and rotten dataset. Its performance may suffer in real-world situations with differing lighting, backgrounds, occlusions, or image noise. Models trained solely on controlled datasets often struggle to generalize to such varying environments.
- **Computational requirements:** While the model can be operated on GPUs such as the NVIDIA RTX 3060, its training and inference demands could introduce challenges in scenarios with limited resources or edge-deployment, such as small-scale farms that do not have access to high-performance hardware.
- **Class imbalance:** Even with the implementation of stratification and augmentation methods, lingering class imbalances in the dataset may influence model fairness and result in biased predictions, particularly for less represented fruit categories. While augmentation and class weighting were effective in mitigating imbalance, advanced approaches such as focal loss and synthetic oversampling (e.g., SMOTE) were not explored in this study and are identified as directions for future work.
- **Dataset differences:** A key limitation of this study is that the evaluation was conducted on a single dataset (Fruits Fresh and Rotten). While this ensured controlled and consistent experimental conditions, it limits the generalizability of our findings. In future work, we plan to perform cross-dataset validation using publicly available benchmarks such as Fruits-360 and FruitNet to better establish the robustness of the proposed hybrid model.

In Table 15 the models were trained using various datasets (e.g., Fruits-360, custom sets). Because metrics can be impacted by these differences in the data set, caution should be used when interpreting the results. A consistent evaluation on the same dataset would make a more direct comparison possible.

5 Conclusion and future work

This research investigated the use of deep learning techniques for the classification of fruit freshness, contrasting pre-trained CNN models (InceptionV3, VGG16, ResNet50, DenseNet121, and EfficientNetB0) with a hybrid CNN-LSTM method. The hybrid CNN-LSTM model effectively captures pseudo-temporal cues from augmented still images, simulating ripening sequences for improved freshness classification. The findings indicated that the hybrid model considerably outperformed the independent pre-trained models, attaining superior accuracy, precision, recall, and F1-score. The addition of an LSTM network with a pre-trained CNN was advantageous in capturing both spatial and temporal characteristics, resulting in a more effective classification system. Moreover, the ablation study underscored the significance of data augmentation and transfer learning in improving model generalization and effectiveness.

These results have important ramifications for the automatic detection of fruit freshness in agriculture and supply chain management. The suggested method can assist in refining harvesting choices, enhancing quality control in supermarkets, and minimizing food waste by guaranteeing precise classification of fruit ripeness. Additionally, the effectiveness of this system renders it a scalable option for extensive deployment in smart agriculture and food logistics.

For subsequent research, multiple avenues can be explored to further improve the efficacy of the proposed model. One possible avenue for enhancement is examining other sophisticated pre-trained architectures to evaluate their influence on classification performance. Furthermore, incorporating multimodal data, such as infrared imaging, hyperspectral data, or spectroscopy, may offer deeper insights into fruit freshness beyond just visual characteristics. Extending this methodology to other perishable items, such as vegetables and dairy products, could also expand the application of deep learning in food quality assessment. While this study focused on apples, bananas, and oranges, future work will incorporate additional fruit types (e.g., grapes, mangoes, papayas) to strengthen model generalizability.

Additionally, model performance could be optimized more effectively than through manual tuning by employing automated hyperparameter optimization techniques like grid search or Bayesian optimization. This would also reduce experimentation time and improve model robustness. Real-time applicability in agricultural environments may be enhanced by investigating lightweight models and edge deployment strategies.

By capitalizing on the advantages of pre-trained models and hybrid deep learning architectures, this research contributes to the development of automated fruit freshness detection systems. Future advancements in AI-driven quality control techniques will persist in playing a crucial role in improving food safety, decreasing waste, and enhancing efficiency in the agricultural industry.

Another important avenue for future work is optimizing the hybrid CNN-LSTM model for real-time deployment on edge devices. Techniques such as model pruning, quantization, and deployment with lightweight CNN backbones will make the framework more suitable for integration into smart agriculture IoT systems.

While the evaluation in this work was primarily based on accuracy-driven metrics (accuracy, precision, recall, and F1-score), future research will focus on domain-specific validation to assess the model's real-world impact. This includes pilot testing with retail outlets or farm supply chains to measure reductions in spoilage-related economic losses

and improvements in shelf-life management. Incorporating such domain-driven evaluations will strengthen the practical relevance of the framework.

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Author Contributions

D.G. conceived the core idea of the hybrid CNN-LSTM model for fruit freshness classification and implemented the experimental framework. J.P.S. contributed to the design of the methodology and supervised the overall research direction. D.G. performed the experiments, analyzed the results, and prepared all figures and tables. D.G. and J.P.S. wrote the main manuscript text. Both authors reviewed and approved the final manuscript

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Data availability

The dataset used in this study, "Fruits Fresh and Rotten for Classification", is publicly available on Kaggle (<https://www.kaggle.com/datasets>).

Declarations

Ethics and Consent to Participate

Not applicable

Consent to Publish

Not applicable

Conflict of interest

The authors declare no Conflict of interest.

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